QBS 103 Final Project - Submission 1

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Submission Notes

Collaborators: none
Additional resources used: course materials; ggplot2 documentation (https://ggplot2.tidyverse.org); data source: Gonzalez-Vazquez et al. / GSE85307, https://pubmed.ncbi.nlm.nih.gov/27841759/ ; google

Choice of gene and covariates

- **Gene:** *LYN* (Affymetrix probe 8146500), an immune-cell signaling kinase.
- **Continuous covariate:** maternal vitamin D (25-hydroxyvitamin D, ng/mL), `mat_vitD`.
- **Categorical covariate 1:** pregnancy outcome (Normal vs. Preeclampsia), `preg_preeclampsia`.
- **Categorical covariate 2 (derived from a continuous variable):** maternal obesity status (BMI $<$ 30 vs. BMI $>=$ 30 kg/mˆ2), created from `mat_BMI`.

Load and wrangle the data

All wrangling is done in R, starting from the original, unmodified files provided on Canvas (the three .csv files must be in the same folder as this .Rmd).

```
library(ggplot2)

# ---- Read the three original files exactly as provided ----
covar <- read.csv("GSE85307_covar.csv", check.names = FALSE, stringsAsFactors = FALSE)
expr  <- read.csv("GSE85307_geneExpression.csv", check.names = FALSE, stringsAsFactors = FALSE)
genes <- read.csv("GSE85307_geneNames.csv", colClasses = "character")

# The first column of the covariate file carries a hidden byte-order mark; rename it cleanly
names(covar)[1] <- "subjectID"

# In the expression file the first (unnamed) column holds the probe IDs: use it as row names
rownames(expr) <- expr[[1]]
expr <- expr[, -1]

# ---- Clean the covariate values ----
# Every covariate value is stored as text such as "age_years: 31.06"; keep only the part after ": "
for (v in names(covar)[3:ncol(covar)]) covar[[v]] <- sub("^[^:]*: ", "", covar[[v]])

# Convert the continuous covariates to numbers
```

```r
num_vars <- c("mat_age_yrs", "delivery_gestAge_wks", "sampleCollection_gestAge_wks",
              "mat_BMI", "mat_vitD")
for (v in num_vars) covar[[v]] <- as.numeric(covar[[v]])

# ---- Match the covariates to the expression columns using geoID ----
# Put the covariate rows in the same order as the columns of the expression matrix
covar <- covar[match(colnames(expr), covar$geoID), ]

# Verify the matching: no sample is missing and every row lines up with its expression column
stopifnot(!any(is.na(covar$geoID)), all(covar$geoID == colnames(expr)))
cat("Samples matched:", nrow(covar), "of", ncol(expr), "expression columns\n")
```

```
## Samples matched: 157 of 157 expression columns
```

```r
# ---- Pull out the gene of interest using the probe-to-gene linker file ----
gene_of_interest <- "LYN"
probe <- genes$PROBEID[genes$GENE == gene_of_interest]
stopifnot(length(probe) == 1, probe %in% rownames(expr))   # exactly one probe, present in the data

# ---- Build the analysis data frame (one row per sample) ----
dat <- data.frame(
  geoID      = covar$geoID,
  expression = as.numeric(expr[probe, ]),        # log2 expression of LYN
  vitD       = covar$mat_vitD,                   # continuous covariate
  outcome    = factor(covar$preg_preeclampsia, levels = c("Normal", "Preeclampsia")),
  BMI        = covar$mat_BMI
)

# Derived categorical variable: obesity status from BMI (obese is BMI >= 30 kg/m^2)
dat$obesity <- factor(ifelse(dat$BMI >= 30, "BMI >= 30", "BMI < 30"),
                      levels = c("BMI < 30", "BMI >= 30"))

# Quick checks without printing the whole data frame
dim(dat)
```

```
## [1] 157   6
```

```r
colSums(is.na(dat))                    # no missing values in the variables used
```

```
##      geoID expression       vitD    outcome        BMI    obesity
##          0          0          0          0          0          0
```

```r
table(dat$outcome, dat$obesity)        # sample size in each boxplot group
```

```
##
##                BMI < 30 BMI >= 30
##   Normal             79        31
##   Preeclampsia       21        26
```

```r
# One shared, clean theme for all three figures (white background, no grid, readable text)
theme_pub <- theme_classic(base_size = 13) +
  theme(plot.title    = element_text(face = "bold", size = 14),
        axis.title    = element_text(face = "bold"),
        axis.text     = element_text(colour = "black"),
        legend.title  = element_text(face = "bold"),
        legend.position = "right")

# Colorblind-friendly colors used consistently across figures
col_main <- "#0072B2"                                          # histogram / scatter
col_obes <- c("BMI < 30" = "#56B4E9", "BMI >= 30" = "#D55E00")      # boxplot groups
```

Figure 1. Histogram of *LYN* gene expression

```r
# Histogram of LYN expression with a dashed line at the mean
ggplot(dat, aes(x = expression)) +
  geom_histogram(bins = 20, fill = col_main, colour = "white") +
  geom_vline(xintercept = mean(dat$expression), linetype = "dashed", linewidth = 0.8) +
  labs(title = "Distribution of LYN expression",
       x = "LYN expression (log2 intensity)",
       y = "Number of samples") +
  theme_pub
```

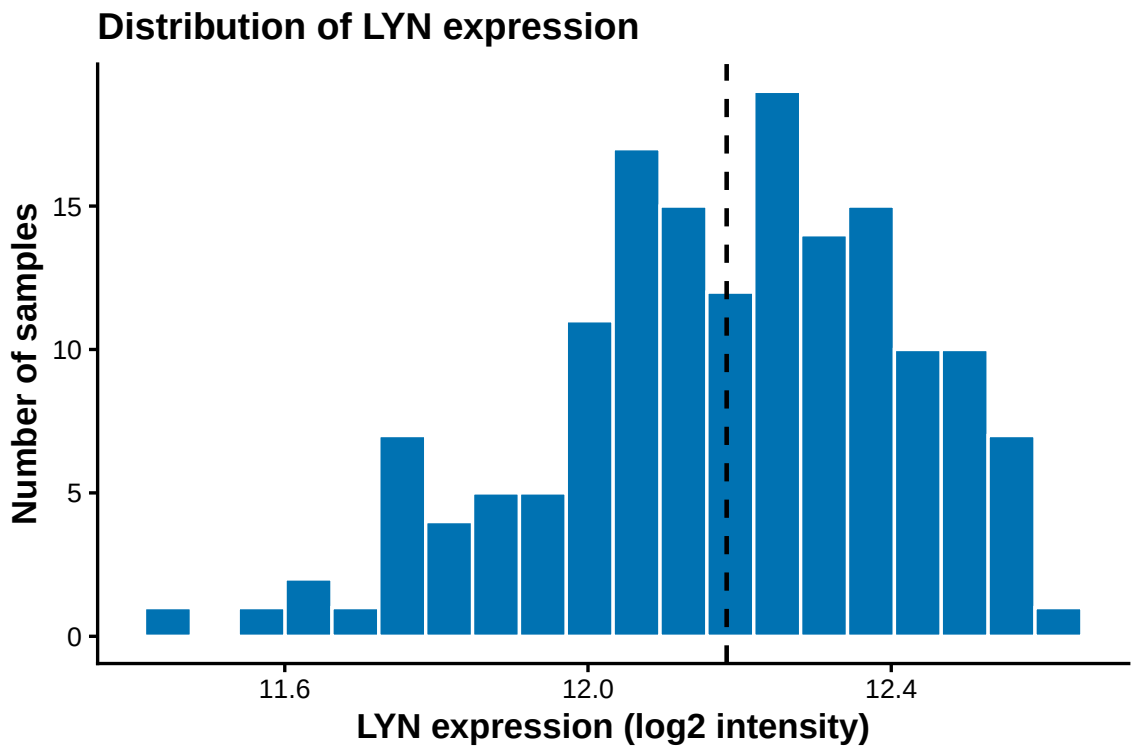


Figure 1. Distribution of *LYN* expression (log2 intensity) in whole-blood samples from 157 pregnant women. The dashed line marks the mean (12.18).

Figure 2. *LYN* expression vs. maternal vitamin D

```r
# Pearson correlation to annotate the plot
ct <- cor.test(dat$vitD, dat$expression)
lab <- sprintf("r = %.2f, p %s", ct$estimate,
               ifelse(ct$p.value < 0.001, "< 0.001", sprintf("= %.3f", ct$p.value)))

# Scatterplot with a linear fit and 95% confidence band
ggplot(dat, aes(x = vitD, y = expression)) +
  geom_point(colour = col_main, alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "black", fill = "grey70") +
  annotate("text", x = min(dat$vitD), y = max(dat$expression), label = lab,
           hjust = 0, vjust = 1, size = 4) +
  labs(title = "LYN expression vs. maternal vitamin D",
       x = "Maternal vitamin D, 25-hydroxyvitamin D (ng/mL)",
       y = "LYN expression (log2 intensity)") +
  theme_pub
```

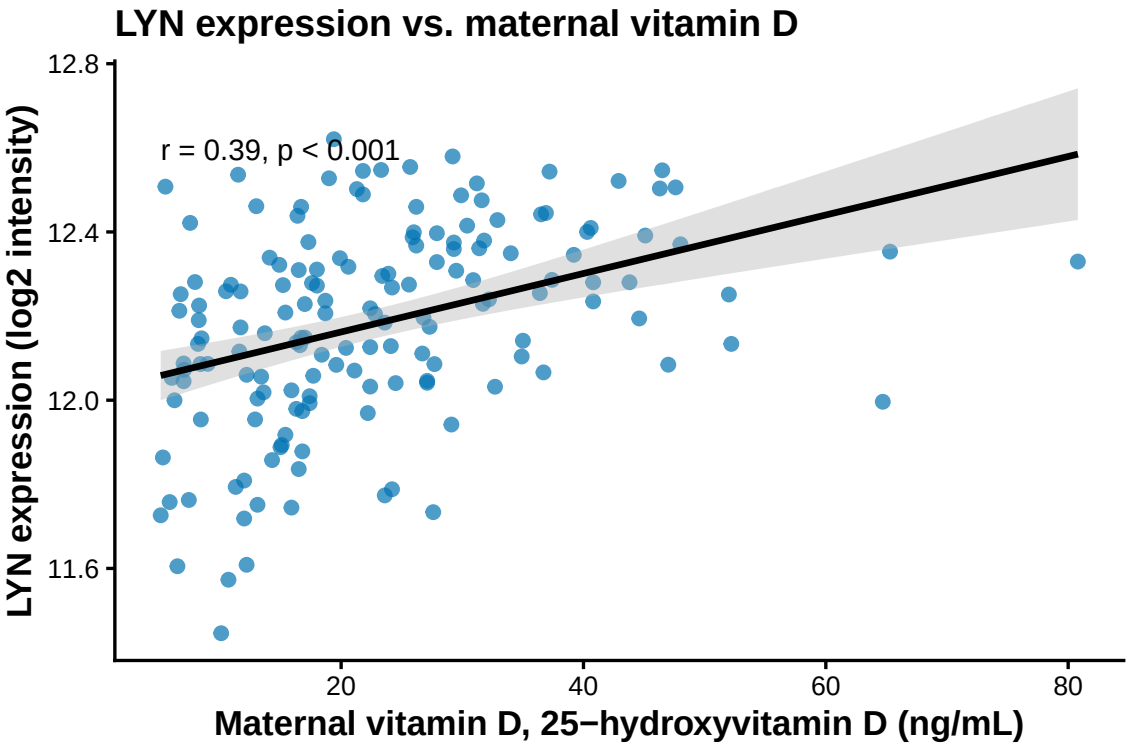


Figure 2. Association between maternal vitamin D (ng/mL) and *LYN* expression (log2 intensity). Each point is one woman; the line is the linear fit with its 95% confidence band ($r = 0.39$, $p < 0.001$).

Figure 3. *LYN* expression by pregnancy outcome and obesity status

```
# Boxplot of LYN expression by pregnancy outcome, split by obesity status (derived from BMI),
# with the individual samples overlaid so the group sizes and spread are visible
ggplot(dat, aes(x = outcome, y = expression, fill = obesity)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8, width = 0.7,
               position = position_dodge(width = 0.8)) +
  geom_point(aes(group = obesity), position = position_jitterdodge(jitter.width = 0.15,
             dodge.width = 0.8), size = 1.4, alpha = 0.6) +
  scale_fill_manual(values = col_obes, name = "Obesity status") +
  labs(title = "LYN expression by pregnancy outcome and obesity status",
       x = "Pregnancy outcome",
       y = "LYN expression (log2 intensity)") +
  theme_pub
```

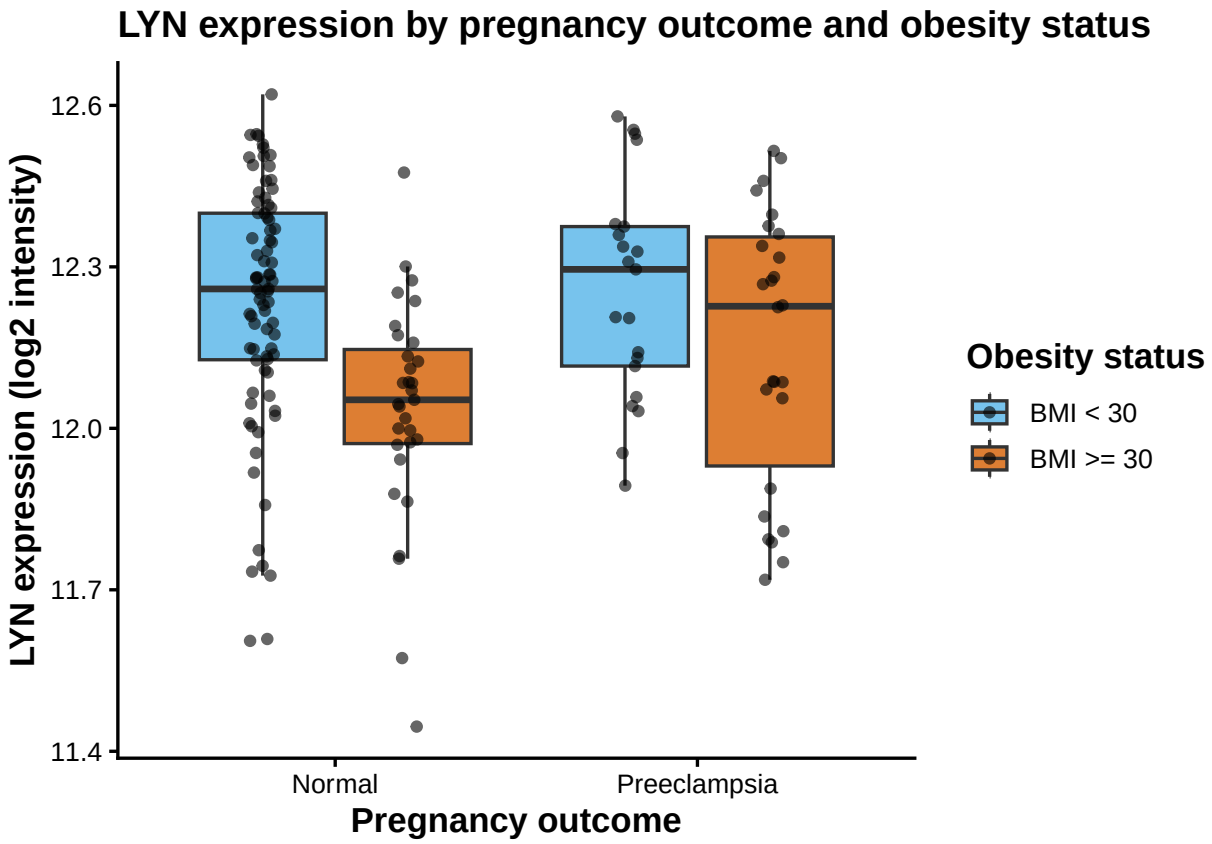


Figure 3. *LYN* expression (log2 intensity) by pregnancy outcome (Normal vs. Preeclampsia) and maternal obesity status (BMI < 30 vs. >= 30 kg/m^2). Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and points are individual samples.